

Project Specification: Student Management & Chat Portal

1. Project Overview

The proposed project is a **Student Management & Chat Portal**. It functions as an educational platform where students can browse available courses, add them to their curriculum (similar to an e-commerce "shopping cart" experience), and communicate with teachers and peers via an integrated chat system.

This application translates the classic "e-commerce" requirements into an educational domain, maintaining the required technical complexity (shopping cart flow using sessions, multiple related database tables, and security) while providing a meaningful, real-world utility.

2. Technology Stack

- **Backend:** Java, Spring Boot, Spring MVC, Spring Data JPA, Spring Security
- **Frontend:** HTML5, CSS3, Thymeleaf (with minimal JavaScript for chat auto-refreshing)
- **Database:** MySQL
- **Architecture:** MVC, Dependency Injection, Session Management

3. Database Entities & Relationships

The project consists of 5 core entities with complex relationships, fulfilling the requirement for relational data modeling:

1. User (Student / Teacher / Admin):

- Stores user details, credentials, and roles.
- *Relationships:* One-to-Many with `Registration`, One-to-Many with `ChatMessage`.

2. Course (The "Product"):

- Represents a course available for enrollment (Title, description, capacity, assigned teacher).
- *Relationships:* One-to-Many with `RegistrationItem`.

3. Registration (The "Order"):

- Represents an official enrollment submission for a semester.
- *Relationships:* Many-to-One with `User`, One-to-Many with `RegistrationItem`.

4. RegistrationItem (The "Order Item"):

- A specific course within a registration request.
- *Relationships:* Many-to-One with `Registration`, Many-to-One with `Course`.

5. ChatMessage:

- Stores chat messages between users.
- *Relationships*: Many-to-One with `User` (Sender) and `User` (Recipient) / `Course` (for group chats).

4. Session Management ("The Cart")

The project fulfills the "ShoppingCart" requirement using a **Course Cart** pattern: When a student browses the course catalog and selects "Add to my schedule", the selected courses are stored in the user's **Session** (Course Cart). The data is only persisted to the database (creating a `Registration` record) when the student finalizes and confirms their enrollment, mirroring an e-commerce checkout process.

5. Key Views and Pages

The application includes the following main views:

1. **Landing / Home Page**: Introduction and platform overview.
2. **Login & Registration Page**: Secured access using Spring Security.
3. **Course Catalog**: A browsable list of available courses with "Add to schedule" functionality.
4. **Registration Cart**: A view of the user's session-stored courses prior to final confirmation.
5. **Student Dashboard & Chat**: The student's personal area showing their enrolled courses alongside an interactive chat interface.
6. **Admin / Teacher Dashboard**: A management interface for creating courses, viewing enrolled students, and interacting via chat.

6. Scope & Requirements Checklist

This project is designed to meet the following expected academic requirements:

- [x] Minimum of 4 inter-related database tables (Has 5).
- [x] Implementation of Session management (Course cart).
- [x] Implementation of Spring Security (Authentication and Role-based authorization).
- [x] CRUD operations on core entities.
- [x] Minimum of 5 distinct web pages/views.